

# Implementation strategy: deploying within 90 days

Bellcourt Review Copilot. PACE stays; the copilot runs beside it.

| When                                      | What happens                                                                                                                                                                                                                                                    | Why / exit test                                                                                                |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Day 0                                     | Send Riverbend the 60-day written notice. Rescind UM-MEMO-2025-19. Re-review the 72 lumbar MRI denials, the 8 Kestrel bariatric denials and the MA glucose-monitor denials. Report timeliness from receipt, not keying.                                         | The notice period is the critical path.<br>The memo fix needs no technology.                                   |
| Days 1 to 30: build and prove             | Host in the Azure tenant that already has a signed agreement for patient data. Connect read-only: fax share, portal webhook, PACE read replica, eligibility files. Load all 38 plan documents into the registry with the account managers. Replay closed cases. | Exit: at least 95% agreement on a fresh, random QA sample of 200 closed cases, zero wrongful approvals.        |
| Days 31 to 60: pilot on self-funded plans | 8 senior nurses and 2 intake coordinators use it on employer queues. Intake sends the drafted fax-back within the hour. Measure minutes per case, agreement and override reasons weekly.                                                                        | ERISA plans are not bound by the state AI laws or the notice clause. Exit: nurse minutes per case down by 25%. |
| Days 61 to 90: Riverbend go-live          | Notice has run. Turn on for Riverbend with Arizona routing and the Texas disclosure in letters. Share audit-log extracts with Riverbend oversight. Train all reviewers.                                                                                         | Exit: MA on-time rate at least 97% and audit accuracy at least 95% for the month.                              |
| After day 90                              | Provider status page and requirements assistant, employer dashboard, policy-change alerts that re-check open cases.                                                                                                                                             | Problems 2 and 3, on the same rule engine.                                                                     |

## Who does what

- **Chief Medical Officer** owns the registry: a policy is not live until its effective dates are in it.
- **Account managers** file every plan amendment into the registry and copy UM Operations.
- **IT (4 people)** provides the read replica, fax share access and Azure hosting. No change to PACE.
- **QA team** audits a monthly sample of tool-assisted cases and owns the go/no-go numbers.
- **Staffing:** fill the 7.4 vacant nurse FTE rather than add 20. Recovering half of the 14.2 minutes spent finding rules is worth 9.3 FTE. License more medical directors in Arizona.

# Design note: technology choices

One page. The stack is deliberately small so it can be explained, audited and replaced.

| Choice            | What I used                                                                                                                                                                  | Why                                                                                                                                                                                                | In production at Bellcourt                                                     |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Pattern           | Agentic RAG with a deterministic rule resolver                                                                                                                               | Plain RAG retrieves by similarity and would return the retired version or the memo. An autonomous agent is not allowed to decide. So: code selects, the model reads and reasons, a person decides. | Same                                                                           |
| Metadata registry | JSON built from each PDF's header plus a small overlay                                                                                                                       | Effective dates, client and authority rank are the missing piece in the UM Library. Filtering on them is what prevents wrong-rule errors.                                                          | Owned by the Clinical Policy Committee                                         |
| Vector store      | 322 section embeddings (text-embedding-3-small, 256 dimensions) in a JSON file, cosine similarity in plain Python, fused with keyword rank                                   | The corpus is 38 documents. A hosted vector database adds a key, a network hop and a failure point with no retrieval gain at this size. The index interface is one function.                       | Azure AI Search or Pinecone with the same metadata filters at 1,100+ documents |
| Model             | google/gemini-2.5-flash, via OpenRouter or the Gemini API directly                                                                                                           | Fast (about 6 seconds a case), cheap (about \$0.004 a case), reads fax images. One environment variable swaps the provider or the model; I tested Claude models too.                               | A model hosted in the Azure tenant under the existing agreement                |
| Agent loop        | Three to five calls: plan retrieval, review, verifier fixes, second check on approvals                                                                                       | Bounded and fully logged. Every step is visible in the audit trail.                                                                                                                                | Same                                                                           |
| Verifier          | Plain code                                                                                                                                                                   | Citations must be in force, quotes must exist in the record, outcome must match the checklist, arithmetic on visit limits is recomputed. The model cannot talk its way past code.                  | Same                                                                           |
| Backend           | Python standard library only, two serverless functions                                                                                                                       | Zero dependencies means nothing to break at deploy time and a tiny attack surface.                                                                                                                 | Azure Functions or a container                                                 |
| Screen and access | One static HTML page. Sign-in with salted password hashes and signed 8-hour sessions; roles enforced on the server                                                           | Case data and fax images are served only to signed-in users. A nurse cannot sign an adverse decision; an auditor cannot run a review.                                                              | Single sign-on with Entra ID, same roles                                       |
| Storage           | Supabase (Postgres) through its REST API, one table for cases, letters and the audit log. Falls back to a local file                                                         | Serverless functions keep no state, so cases, decisions and letters need a database. One generic table keeps the prototype simple.                                                                 | Azure SQL or Postgres inside the tenant, with proper tables                    |
| Letters and email | Letter text built in code from the human decision and the cited provision. A small built-in PDF writer. Gmail SMTP for the demo, delivered only to a configured demo mailbox | A letter is never generated by the model alone and never sent without a human decision. No extra dependency.                                                                                       | Bellcourt's correspondence system and print vendor                             |
| Evidence          | Scripts that write JSON and the PDFs from the same run                                                                                                                       | Numbers in the documents cannot drift from the numbers in the app.                                                                                                                                 | Monthly QA job                                                                 |

**What I would change with more time:** a database-backed audit log instead of browser storage; a fresh held-out audit sample; a stronger model for the review step if the fresh sample shows drift; plan documents for the other 32 employers.